BERN03 Exercise 1: Handling climate model output I

In this exercise we will use MATLAB to analyse output from one of the model simulations in the study of Jungclaus et al. (2010). The simulation is called 'mil0021' and you can download model output for seasonal means (JJA, DJF) of 2m temperature (TEMP2), mean sea level pressure (MSLP) and total precipitation (TPREC), as well as some simple MATLAB scripts using the following link: https://lu.box.com/s/8qowhiznk514d8dwqyx7j1ipue6b43op

Using the mil0021 model simulation and MATLAB, do the exercises below. Take notes and comment your code, so that you at a later point understand what you have done!

1. Load the model output in MATLAB using the function 'netcdf_simple.m', like this:
   [matlab_variable_name] = netcdf_simple('file_name','model_variable');
   The 'model_variable' names in the NetCDF files are '2t', 'slp', and 'precip' for TEMP2, MSLP and TPREC, respectively. Also load the 'lon' and 'lat' matrices from the file 'lon_lat_mat.mat'. Have a look at the variables in the MATLAB workspace. Check the dimensions etc. NOTE: the functions used in this exercise don't all assume the same orientation of the model data. Check if you get error and use 'permute.m' to reshuffle the dimensions of the data.
2. Plot maps of seasonal mean using the MATLAB function 'surfm'. Add continents using the MATLAB coastline data.
3. Using the function 'site_interp_echamT31.m' extract timeseries of DJF MSLP at the NAO centers of action Reykjavik, Iceland (64.15°N, -21.94°E) and Ponta Delgada, Azores (37.74°N, -25.68°E). Then calculate the NAO index as the normalized pressure difference between the two timeseries. Plot the simulated NAO. How would you describe the data?
4. Using the 'Rean_regr.m' function to calculate the correlation, plot correlation maps of the NAO correlation to DJF MSLP, TEMP and TPREC. How does the maps correspond to what we know about the NAO?
5. Using the function 'area_mean_model.m' canculate the temperature series for ENSO 3.4 (area: 5°N-5°S, 120°W-170°W). Perform similar analysis to the one for NAO. Look at the ENSO timeseries and plot correlation maps. How does your analysis correspond to what we know about the ENSO?
6. Make a timeseries of the Indian Summer Monsoon (ISM). From the model data, and the approach you have learned above, try to understand some of the dynamics governing the ISM.